

Ali Ahmadi Esfidi

AI Researcher & Developer

✉ aliahmadiesfidi@outlook.com | 🌐 Mr-Ahmadi | 🌐 Ali Ahmadi Esfidi | 🌐 Portfolio

Education

B.Sc. Computer Science > Amirkabir University of Technology 📅 Sep 2022 – Sep 2026

• Location: Tehran, Iran • Field of study: Computer Science • GPA: 18.93/20 • Top 20%

High School Diploma > Dr. Shahriari High School (SAMPAD) 📅 Sep 2019 – Jun 2022

• Location: Qom, Iran • Field of study: Mathematics & Physics • GPA: 18.42/20

Research Interests

- Deep Reinforcement Learning
- Machine Learning for Decision-Making Systems
- AI for Infrastructure Monitoring
- Optimization Methods for AI and Networking
- Computational Biology & Bioinformatics
- Software-Defined Networks & Resource Allocation

Experiences

Research Assistant > Supervisor: Prof. Ghatte 📅 Jan 2025 – Present

📍 NORC Lab, Amirkabir University of Technology

- **Led** an AI team and **engineered** web systems and deep learning models for automated bridge damage detection in collaboration with the Tehran Urban Planning and Research Center.
- **Gained hands-on experience** in developing efficient deep-learning pipelines for computer vision applications, focusing on model compactness and structural health monitoring.
- *Stack:* **PyTorch** • **OpenCV** • **ReactJS** • **NodeJS** • **PostgreSQL** • **Docker**

Research Assistant > Supervisor: Prof. Khonsari 📅 Sep 2024 – Present

📍 High Performance Networks Lab, Tehran University

- **Developed** deep reinforcement learning systems for complex agricultural decision-making, focusing on reward engineering and policy optimization to balance multiple objectives.
- **Acquired proficiency** in benchmarking reinforcement learning algorithms across discrete and continuous action spaces, and in utilizing simulation environments for training and evaluation.
- *Stack:* **PyTorch** • **TensorFlow** • **Pandas** • **NumPy**

Research Assistant > Supervisor: Dr. Dolati 📅 Sep 2025 – Present

📍 Sharif University of Technology

- **Formulated** MDPs for switch pairing and collaborative rule caching, and designed simulation environments for joint scheduling-routing policies in programmable data planes.
- **Implemented** optimization frameworks for in-network aggregation in distributed machine learning, focusing on ILP decomposition, flexible routing strategies, and scalable algorithmic design.
- *Stack:* **PyTorch** • **Numba** • **Pandas** • **NumPy**

Teaching Assistant > Graduate & Undergraduate Courses 📅 Sep 2024 – Present

📍 Amirkabir University of Technology & Tehran University

• Data Mining B.Sc.	Dr. Yousefimehr	Sep 2026 – Present
• Quantum Information Processing M.Sc.	Prof. Khonsari	Sep 2025 – Jan 2026
• Computational Data Mining M.Sc.	Prof. Ghatte	Sep 2025 – Jan 2026
• Introduction to Logic B.Sc.	Dr. Didehvar	Sep 2025 – Jan 2026
• Theory of Computation B.Sc.	Dr. Didehvar	Sep 2024 – Jun 2026
• Artificial Intelligence & Workshop B.Sc.	Prof. Ghatte, Dr. Yousefimehr	Sep 2024 – Jan 2025

Publications

- M. Saberi, A. Ahmadi Esfidi, M. Dolati, N. Omidvar, **FlexINA: In-Network Aggregation for Accelerating Distributed Machine Learning with Flexible Routing**, Accepted for publication in *IEEE Transactions on Network and Service Management*. [View Paper](#).

- P. Heidari, A. Khonsari, [A. Ahmadi Esfidi](#), A. Mehrvarz, E. Khodaei, A. Dadlani, **DeepIrrigo: Deep Reinforcement Learning for Continuous and Discrete Action Irrigation Optimization — A Case Study in Iran**, Under review at *Intelligent Systems with Applications*. [View Paper](#).
- M. Saberi, [A. Ahmadi Esfidi](#), M. Dolati, A. Khonsari, A. Movaghar, **Rule Caching in Programmable Networks with Deep Reinforcement Learning**, Submitted to *IEEE Transactions on Network and Service Management*.
- P. Heidari, [A. Ahmadi Esfidi](#), A. Mehrvarz, E. Khodaei, A. Khonsari, **Irrigation Optimization in Agricultural Fields Using Deep Reinforcement Learning Approaches**, Published in *CSICC 2025*. [View Paper](#).

Selected Projects

 [See more](#)

DRL-Based Beamforming Optimization for Movable Intelligent Surfaces

 Jun 2026 – Present

- **Designed** a DRL framework to optimize beamforming for MIS-assisted SWIPT systems, jointly configuring phase shifts and power splitting ratios to maximize energy harvesting.
- *Stack:* [PyTorch](#) • [Gymnasium](#) • [MOSEK](#) • [NumPy](#)

RL-Based ML Job Scheduler with Distribution Optimization

 Aug 2025 – Oct 2025

- **Developed** a two-stage hierarchical PPO system for scheduling ML jobs on multi-server accelerator clusters using the Gavel dataset, reducing average job completion time.
- *Stack:* [PyTorch](#) • [Gymnasium](#) • [NumPy](#)

Machine Learning Perspectives on Gold-Binding Peptides

 May 2025 – Jun 2025

- **Applied** machine learning to classify and predict gold-binding affinity of peptide sequences, comparing regression and classification approaches across multiple featurization strategies.
- *Stack:* [PyTorch](#) • [Scikit-learn](#) • [Hugging Face](#) • [Pandas](#) • [NumPy/SciPy](#)

RNA Pairing Pattern Recognition Model

 Mar 2024 – Jul 2024

- **Developed** a system to predict RNA secondary structures using SCFGs, evolutionary information, and enhanced CYK parsing for complex motifs; presented at [CBRC Journal Club](#).
- *Stack:* [NetworkX](#) • [Biopython](#) • [PhyML](#) • [NumPy/SciPy](#)

Skills

Programming Languages	Python • JavaScript • TypeScript • R • Java • C++ • Swift
Frameworks	React • Node.js • Express.js • Django • Electron
Data Science / ML	PyTorch • TensorFlow • Pandas • Scikit-learn • NumPy • OpenCV • LangChain
Platforms & Tools	Docker • Git • GitHub • MongoDB • PostgreSQL • Hugging Face

Certifications

Bioinformatics Internship Program > [Biocan](#)

 Nov 2025

Scientific Secretary: Dr. Z. Salehi • Course Director: Dr. K. Kavousi

Introduction to Bioinformatics > [Biocan](#)

 Jul 2025

Scientific Chair: Dr. K. Kavousi • Score: 96/100 • Top 3% of 350+ students

Scrum Foundations Course > [Ultima Training Tech](#)

 Dec 2024

Instructor: Josef Balahan

Volunteering

Student Scientific Magazine “Halghe” > Amirkabir University of Technology

 Dec 2025 – Present

Contributed to scientific editing and supported the editorial team

Technical & Web Support Volunteer > Amirkabir University of Technology

 May 2023, Feb 2025

Web and procurement support for DAI DAY (2025) and the International Conference on AI and Smart Vehicles (2023)